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Exposing deepfake using fusion of deep-learned and hand-crafted features

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Abstract
With the milliards of images and videos, visual content has become a critical source of information. The spread of misinformation has become a significant problem through the availability of editing tools, requiring robust manipulation detection methods. Some of the manipulations, such as copy-move and splicing, are easy to detect, while other advanced facial manipulations, such as DeepFake are hard to detect. Facial manipulations can change human expressions by creating highly realistic faces. In this paper, we propose an efficient method to expose DeepFake in digital videos. A fusion of hand-crafted and deep-learned features is utilized to improve detection performance. The image quality measure (FM) is used besides the similarity measure of face and body skin color to generate the hand-crafted features. The experimental results show the efficiency of the proposed method for exposing DeepFake. We conducted the experiments on the three commonly and publicly available datasets Celeb-df, DFDC, and Faceforensics++.

Keywords Face forensics · Video manipulation detection · Convolution neural network · Facial manipulation detection · Frequency domain · Blurred images

1 Introduction

Nowadays, DeepFake images and videos invade social media networks such as Facebook and Twitter, destabilizing confidence in visual content. DeepFake refers to the manufacturing of any facial manipulations [1, 23, 28] such as expressions, identity, age,...etc. With the

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availability of editing tools, it becomes effortless to manipulate visual content, not only by professionals but also by anyone. These tools have contributed to the advancement of society, especially in the field of advertising and the film industry. However, they have been misused to distort some people like world leaders, celebrities, and politicians and create fake news [4], leading to chaos and general instability. Moreover, some of the outrageous use cases of DeepFakes have been pornography, often featuring females.

Visual manipulation is considered harmful or forged if the content has been distorted for a deception that can easily influence people's decisions and thoughts through fabricated content. Therefore, verifying visual content is vital to preserve the community and make it more stable.

Digital visual manipulations can be divided into three categories: image manipulation (IM), video manipulation (VM), and facial manipulation (FCM). In the first category, IM, any part or object in the image is manipulated, such as splicing and copy-move [10, 12, 35, 46]. In splicing, a portion of an image is copied and pasted into another image. During the copy-move, a part of an image is copied and pasted into another position in the same image. VM refers to manipulation in video content, whether in the spatial, temporal, or spatiotemporal domain, such as intra-frame forgery and inter-frame forgery [2, 8, 11]. In FCM, just a human face is subjected to manipulation, and the rest of the content is preserved [23, 38, 47]. Recently, the term DeepFake has been used to denote any FCM [19] such as Face2Face, FaceSwap, NeuralTextures,...etc.

FCM is considered a high-risk level because the human face plays a central role in the emergence of interactions confirming or conveying a specific message, where any manipulation leads to establishing a wrong message. In the last three years, with the rapid advancement of deep learning, FCM generation and detection have attracted significant attention from researchers worldwide, incredibly. The development of deep learning and the ability to generate high-fidelity images using variants of generative adversarial networks (GAN) [17] have made it possible for anyone to generate a realistic-looking face whose identity does not exist in the world or to perform realistic facial manipulations, such as identity swapping, in a video. The latter is what people in the know call the DeepFake. The issue is that high-performance DeepFake generating models are widely available and have low entry barriers. When employed maliciously, DeepFake poses a severe and immediate danger to the integrity of the news we receive.

The degree of danger associated with various DeepFake operations varies. For instance, identity swap, which can change the subject's hard biometrics [18], is the most dangerous, followed by expression swap and attribute manipulation, which only tamper with the subject's soft biometrics [18]. Finally, whole-face synthesis, which is not based on manipulating the subject's hard or soft biometrics, is the least dangerous. Figure 1 shows some examples of DeepFake, where it is not easy to differentiate between fake and real faces.

In addition, a lot of available face editing tools (i.e., Face App, FaceTune2, Zao, VSCO, etc.) are capable of modifying facial expressions, which have received wide attention in the social environment in many countries [37], leading to questionable trust in visual content and digital communications. Therefore, validating visual content is an urgent task to verify the message and impression. Although significant efforts have been made to detect such manipulations, the detection performance is still unsatisfactory, coinciding with the advancement of generation methods.



Fig. 1 Examples of real and DeepFake manipulation from Celeb-df dataset [26]

In this paper, we focus on exposing DeepFake by proposing a novel method that fuses deep-learned and hand-crafted features, which achieves high detection performance. The main contributions of our work can be summarized as follows:

- We propose a novel DeepFake-FCM detection system using a fusion of deep-learned and hand-crafted features, achieving superior results compared to the previous detection methods.
- We propose an efficient hand-crafted feature extraction method that depends on image quality measure (FM) for detecting the amount of blurriness in the frequency domain.
- We use a pre-trained VGG network as a feature extractor, which takes advantage of CNN to extract the deep features.
- We apply feature fusion to produce a feature vector of each image, improving the classification accuracy.
- RBF-MSVM is used as a classifier to improve the detection accuracy of the proposed system.
- We conduct various experiments on different datasets to evaluate and compare the proposed method with state-of-the-art methods.

The rest of this paper is organized as follows: Section 2 briefly reviews the existing methods in the literature, followed by the motivation section in 3. Section 4 discusses the manipulation artifacts that have been observed. Section 5 discusses the proposed system for detecting DeepFake. Experimental results and discussion are given in Section 6. Finally, the conclusion is given in Section 8.

2 Related works

DeepFake Databases. Recently, there have been amazing developments in the generation of synthetic media that rely on artificial intelligence. Based on deep learning approaches, generating visual media with a high level of realism is now possible.

Various FCM generation algorithms (i.e., FaceSwap [22], NeuralTextures [16], Face2Face [42], and FSGAN [33]) have been introduced to create synthesized face images that seem as original faces. These algorithms allowed the research community to grow FCM benchmarks for public usage such as Celeb-DF [26], Faceforensics++ [38], (DFDC) [9], etc.

Large-scale challenging DeepFake videos have been presented in the Celeb-DF dataset. It contains high-quality DeepFake videos of synthesized faces of celebrities. FaceForensics++ is considered a famous FCM benchmark consisting of 1000 real videos that have undergone manipulation with four methods: DeepFakes, Face2Face, FaceSwap, and NeuralTextures.

Facebook has introduced a large DeepFake dataset called DFDC. It contains 128,514 videos, including over 100,000 clips sourced from 3,426 actors. Various face swap techniques have been utilized with the distractor and augmenter. Distractor indicates object overlay (i.e., face, shapes, and text), while augmenter indicates geometric and color transforms, frame rate changes, etc.

Furthermore, the VidTIMIT dataset collected by Korshunov and Marcel [21] used a GAN-based approach to create swapped faces. Prajwal et al. proposed a new model called Wav2Lips [34] that introduces a lip-synchronization network for generating lip-syncing for arbitrary talking-face videos.

Detection of DeepFake. In recent years, numerous studies have continually developed various approaches to determining whether a video is synthesized with AI (GANs) or genuine. This identification process has been ongoing, and the ability of a detector approach to identify unseen synthesized artifacts is resilient against diverse attacks (such as image/video modifications). The detection approaches can be categorized into hand-crafted-based methods and learning-based methods.

For hand-crafted-based methods, Fridrich and Kodovsky [13] used co-occurrence patterns as hand-crafted features to detect FCM. Matern et al. [28] extracted the texture features [24] of the nose area and face border. Then a neural network was used for classification. Wang et al. [43] utilized the co-motion pattern, which is the temporal motion of multiple specific spatial locations. In Yang et al. [45], 3D head poses were estimated from the face images to observe errors that could be revealed. An SVM classifier was used for training and detecting DeepFake. They also proposed a DeepFake detection method [25] by analyzing the face-warping artifacts. Li et al. [27] utilized chrominance components in the residual domain to distinguish between synthesized and real faces. They built a feature set to capture color image statistics and a model for identifying real and fake images. McCloskey and Albright [29] trained an SVM-based classifier to identify fake faces depending on measuring the frequency of saturated and under-exposed pixels in each facial image. Megahed et al. [31] developed a model that identified Face2Face modifications by combining HOG characteristics and a liner-SVM algorithm. These previous methods rely on barely observed artifacts, but their detection performance reduces when such artifacts become more challenging to obtain. According to the generated videos in recent benchmarks, they are more realistic, and many manipulation artifacts have vanished. Suganthi et al. [41] introduced a method for detecting deepfake face images using the deep learning technique of fisherface using Local Binary Pattern Histogram

(FF-LBPH). Then Deep Belief Network (DBN) with Restricted Boltzmann Machine (RBM) was applied for classification.

Additionally, several methods have been proposed to detect videos containing FCM based on deep learning. Cozzolino et al. [6] utilized the previous steganalysis features presented in [28] and fed them to the CNN network. This network has been fine-tuned and evaluated on the Faceforenscis++ dataset in [38]. Bayar and Stamm [3] proposed a CNN network containing eight layers and trained it on the Faceforenscis++ dataset. Rahmouni et al. [36] used various architectures of CNN with a global pooling layer (i.e., maximum, minimum, mean, and variance). Zhou et al. [47] used a two-stream network and combined the output scores. Where the first stream was used to detect low-level inconsistencies among image patches, and the second stream was used to detect the tampered faces. Afchar et al. [1] proposed two CNN architectures based on agnostic to the technique of FCM. Kumar et al. [23] split the face images into four parts, and each part was fed in parallel to ResNet-18 models, and the outputs were concatenated as one vector for classification. Sabir et al. [39] used a combination of recurrent convolutional models to detect DeepFake. In Fung et al. [14], a novel DeepFake detection method was proposed using an unsupervised learning technique. First, two different transformed versions of an image were generated and fed into an encoder and a projection head networks sequentially. Khalid et al. [20] introduced a model called OC-FakeDect. They utilized a variational autoencoder as an anomaly detection issue. Wu et al. [44] used a recurrent network to extract temporal features for detecting inconsistency between consecutive frames. Combining motion and appearance features, Megahed et al. [32] used a volume local binary pattern (VLBP) to generate dynamic texture images from each video clip. To further identify deepfake and retrieve learned manipulation artifacts, they proposed a CNN network called FMD-Net. Fung et al. [15] presented a novel unsupervised learning approach to identifying deepfake. Face images are modified twice, once using Xception and once using a sequential subnetwork (the projection head network). In addition, the projection head networks' outputs improve the agreement to the greatest extent possible.

3 Motivation

Visual contents, particularly those that show the presence and actions of people, are often taken as proof that the events happened. Recent technologies make it easier and more available to change these kinds of content, which helps to change people's thinking. When fake news circulates on social media and is believed by people who lack the tools to determine whether it is true or not, this counterfeiting puts society at risk. Therefore, knowing how to expose fake videos is becoming more critical. Some noticeable manipulation artifacts in fake videos can be utilized to differentiate between fake and real.

However, there are existing methods based on hand-crafted features to detect visible artifacts in specific regions (i.e., eyes, noses, etc.), but such methods fail with invisible artifacts. On the other hand, deep-learning-based methods claimed success in detecting DeepFakes with high confidence. Most failed to evaluate the effectiveness of handling undetected DeepFakes and the detector's robustness against image/video transformations (such as blurring and compression) and still need improvements.

This paper combines hand-crafted and deep-learned features to detect DeepFake and improve performance. For hand-crafted features, FM is used as an image quality measure for identifying an amount of blurriness in the frequency domain. Moreover, the similarity

measure of the color features between the face region and the rest of the skin regions is computed to detect color and illumination artifacts.

In recent years, CCN networks have been successful in graph structure data representation and learning, which have demonstrated excellent performance in many computer vision applications such as object tracking, semantic segmentation, and image recognition. Inspired by these breakthroughs, we utilized a pre-trained VGG model to detect manipulation artifacts and inconsistency in fake videos, which achieved improved results.

4 Preliminaries

Nowadays, using deep learning networks has become prominent and widely used in various applications. Facial manipulations have gained significant momentum with this development, and advanced manipulation methods have appeared. Therefore, it became visually hard to differentiate between synthetic and real faces. After analyzing various facial manipulation benchmarks, we observed that some artifacts (i.e., obliteration, blurriness, illumination changes) were spotted in the synthesized faces and faded in the real faces. These artifacts can be exploited to detect FCM and expose Deepfake.

FCM manipulation artifacts. Among the most prominent artifacts observed was the obliteration of details in some regions, such as teeth and eyes. Figure 2 shows an example of detail obliteration where the details of the mouth and eye are clarified in the real face (Fig. 2a). At the same time, these details are obliterated in the fake face (Fig. 2b). Another observed artifact is the blurriness, which increases in the fake face as shown in Fig. 3.

On the other hand, there is a difference in the color intensity and illumination between the skin of the manipulated face region and the remaining body skin, as shown in Fig. 4.

Deep learning-based models are widely used in various fields, such as image classification, image and video forensics, face recognition, etc. Numerous pre-trained deep learning models have become available to extract features automatically, making rapid progress in the past few years.

According to the previous artifacts, deep-learned features extracted from the pre-trained VGG model are utilized to exhibit and identify DeepFake. To improve the performance of our detection system, we propose a fusion method that fuses hand-crafted features with deep-learned features. In hand-crafted features, the image quality measure [7] is used for each face region and its parts (i.e., left_eye, right_eye, and mouth) to compute the amount of blurriness in the frequency domain. Moreover, a correlation between the color features of the face region skin and the remaining body skin [30] is computed.



(a) Real Face (b) Fake Face

Fig. 2 Example of details obliteration (Celeb-df [26])



Fig. 3 Example of image blurriness in real and fake images (Celeb-df [26])

5 Methodology

This work introduces a novel method to automatically and efficiently detect DeepFake by fusing deep-learned and hand-crafted features. Figure 5 shows the block diagram of the proposed method, which consists of four main stages: pre-processing, feature extraction (i.e., CNN feature extraction, FM feature extraction), feature fusion, and classification. The pre-trained VGG model is used as a deep feature extractor from each face image. While the

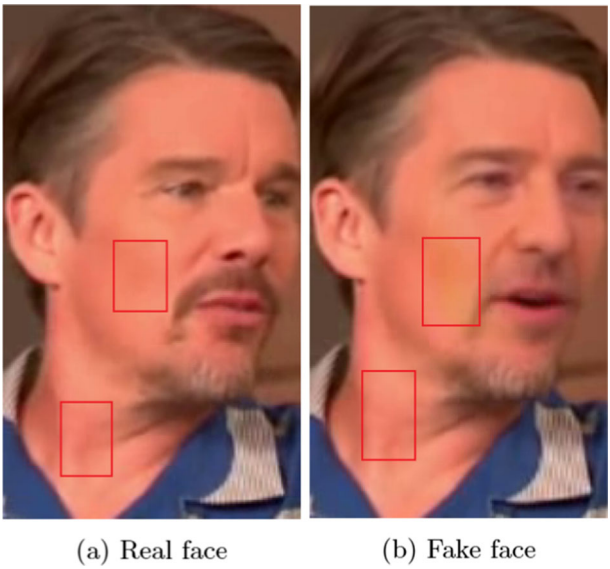


Fig. 4 Example of color and illumination artifacts (Celeb-df [26])

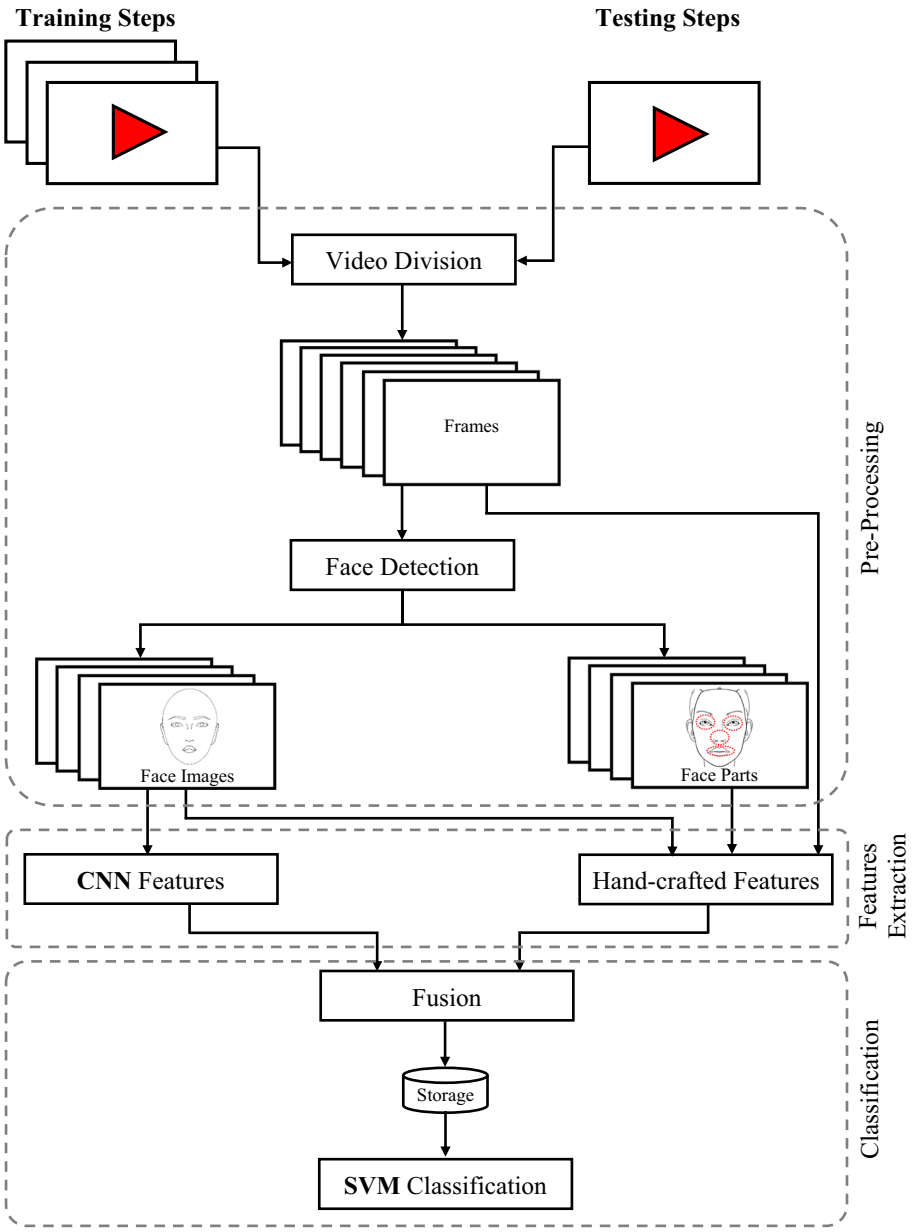


Fig. 5 General block diagram of the proposed method

FM algorithm is used as a hand-crafted feature extractor from a face image, the similarity measure between color features (i.e., mean, standard deviation, and skewness) is computed. Then, these features are fused to form a single feature vector for each image. Finally, an SVM classifier is used to classify each image as real or fake. The following subsections describe each aspect of the proposed method in more detail.

5.1 Pre-processing

Firstly, the pre-processing step applies to each input video to prepare it before feature extraction. An input video is split into small non-overlapping shots (i.e., 10 frames per shot). One frame is randomly selected from each shot, and the rest frames are ignored. For each selected frame, a face detector [1] is applied to detect and crop the face region and all face parts regions (i.e., left-eye, right-eye, nose, and mouth) rather than using full frames as shown in Fig. 6. Each face part image will be used in the feature extraction step.

5.2 Feature extraction

To compute the whole feature vector of each face image, we utilize a deep learning model to extract learned features, besides some hand-crafted features, based on the amount of blurriness and skin color features.

5.2.1 Deep-learning feature extraction

A pre-trained VGG-m model is used for feature extraction. VGG is a model used for image recognition tasks proposed by the visual geometry group at the University of Oxford [5]. It is utilized to extract deep features from the face images by the convolutional base of the network, and the rest of the network is discarded. The architecture of VGG-m is described in Fig. 7; It consists of 11 layers (5 convolution layers, 3 max-pooling layers, and 3 fully-connected layers). RELU is used as an activation function of all layers in the convolutional base. This model is fine-tuned on the Celeb-df [26], DFDC [9], and Faceforensics++ [38] datasets to adapt the specialized features and increase the detection performance. All face images are resized to $224 \times 224 \times 3$ as the model input. We extract the deep-learned feature vector of each face image from the first fully connected layer (fc6), where the feature vector size is 4096.

5.2.2 Hand-crafted feature extraction

a. Amount of blurriness

Counterfeiters usually resort to blurring the manipulated face to hide the manipulated artifacts. When the facial area deteriorates with excessive blur, detecting it becomes challenging. To compute the amount of blurriness in the image of the face and its parts (left_eye,right_eye, and mouth), we use FM proposed by Kanjar and Masilamani [7]. They proposed an image quality measure in the frequency domain using Fourier transform. When the amount of blurriness increases, the FM value decreases. First, the Fourier transform is computed as:

$$Y_{p,q} = \sum_{i=1}^{M} \sum_{j=1}^{N} \omega_M^{ip} \omega_N^{jq} X_{i,j} \tag{1}$$

[1] Masayuki Tanaka (2021). Face Parts Detection https://www.mathworks.com/matlabcentral/fileexchange/36855-face-parts-detection, MATLAB Central File Exchange. Retrieved September 4, 2021.

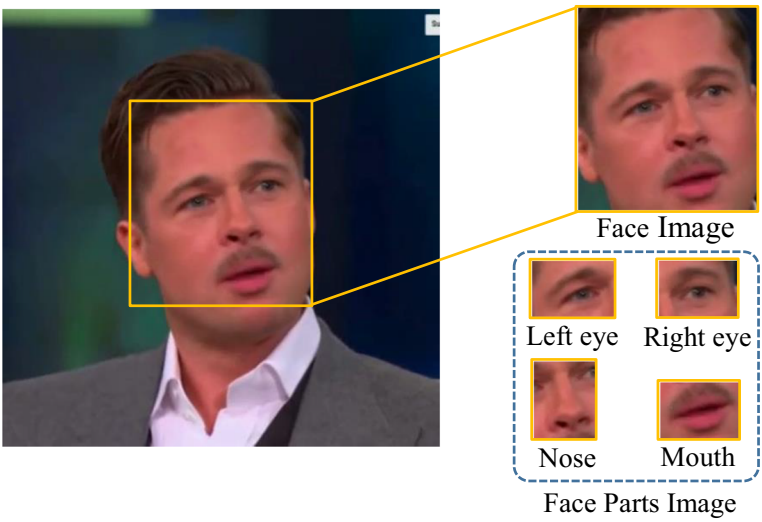


Fig. 6 An example of face parts detection; five images (Face, left_eye, right_eye, nose, and mouth) as output of the face detector

where ω_m and ω_m are complex roots of unity as:

$$\omega_M = e^{-2\pi k/M} \tag{2}$$

$$\omega_N = e^{-2\pi k/N} \tag{3}$$

Where k is the imaginary unit, then Fc shifts the indices for X and Y by 1 to reflect matrix indices. FM is computed according to the number of high-frequency values. When the amount of blur increases, the number of pixels with values greater than the threshold decreases. The procedure for calculating FM is summarized in Algorithm 1.

b. Color features of skin

In addition, a skin detection technique [40] using YCbCr color space is applied to extract all of the body's skin from the original frame (Fig. 8b). We use face image indicators to separate face skin from all body skin, as shown in Fig. 8c and d.

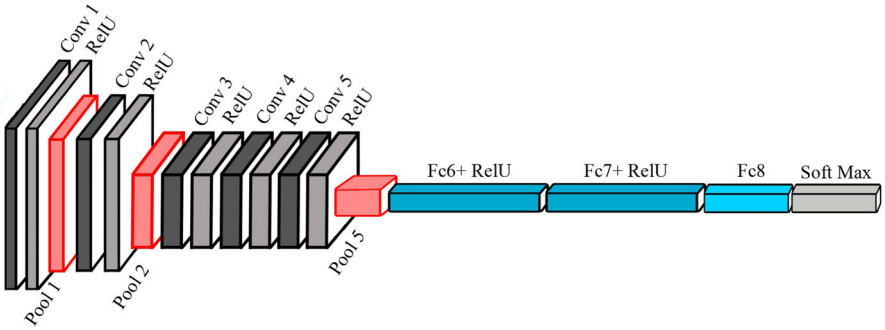


Fig. 7 Architecture of the VGG-m network

Algorithm 1 Procedure for FM computation.

```
I ← Input Image of size m × n
F ← Fourier Transform of I
Fc ← F shifting
M = max(Fc)
thres = M/1000
T_H ← intial value 1
for i = 1, 2, ...length(F) do
    if F(i) > thres then
        T_H ← T_H + 1
    end if
end for
FM = T_H/(m × n)
Return FM
```

After that, the three-color features [30] (i.e., mean, standard deviation, and skewness) are computed from the face skin and the remaining body skin as the following equations:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} P_i \tag{4}$$



Fig. 8 Illustration of skin detection

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}|P_i - \mu|^2} \tag{5}$$

$$SK = \frac{\left(\frac{1}{N}\sum_{i=1}^{N}|P_i - \mu|\right)^3}{\sigma^3} \tag{6}$$

Where N is the number of pixels and P_i is the pixel's intensity value. Therefore, we have two feature vectors: $C_f = \{\mu, \sigma, SK\}$ and $C_b = \{\mu, \sigma, SK\}$ (i.e., the color feature vector of face skin and for body skin). To compute the similarity measure between the features of face and body skin, the correlation is calculated as follows:

$$cor\left(C_f, C_b\right) = \frac{\sum\left(C_f - \mu_f\right)\left(C_b - \mu_b\right)}{\sqrt{\sum\left(C_f - \mu_f\right)^2}\sqrt{\sum\left(C_b - \mu_b\right)^2}} \tag{7}$$

Where C_f, C_b are the color feature vectors of face and body, and μ_f, μ_b are mean of C_f, C_b respectively.

5.2.3 Fusion

After applying the FM measure, we have four feature values: FM_face, FM_left eye, FM_right eye, and FM_mouth. The fifth feature value in the hand-crafted feature vector is *cor*. Figure 9 shows the five hand-crafted features.

Finally, the deep-learned features vector and the hand-crafted features vector are fused by concatenating two vectors, where the final fusion feature vector length is 4101.

5.3 Classification

Support vector machine (SVM) is applied for binary classification that classifies the face image as real or fake using a Gaussian radial basis kernel function as

$$G\left(y_i, y_j\right) = y_i' y_j \tag{8}$$

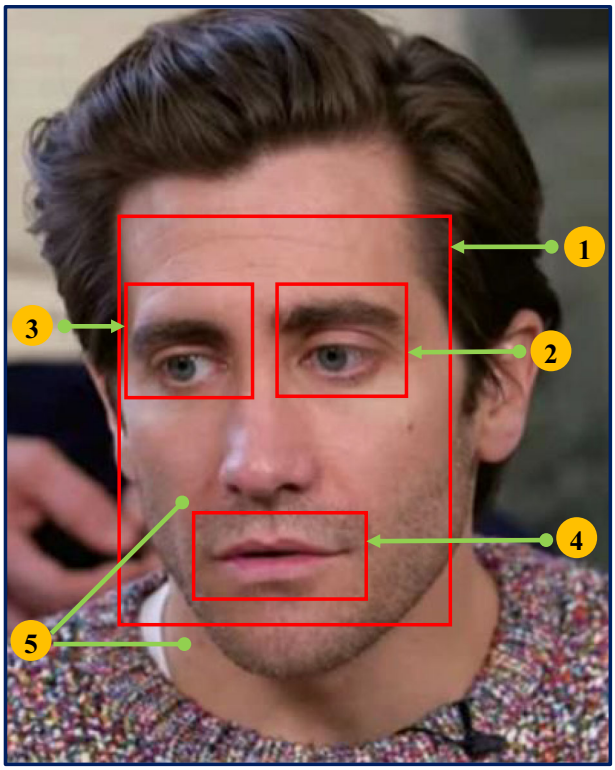
Kernel function is used to compute the elements of the Gram matrix. Assume $G(y_i, y_j)$ is the element (i, j) of the Gram matrix, where y_i and y_j are p-dimensional vectors representing observations i and j in y. Figure 10 illustrates a workflow diagram with all the proposed method procedures.

6 Experimental results

6.1 Experimental setup

The proposed system has been trained on an Ubuntu 18.04 LTS workstation with an Intel Core i9-9820X Skylake X 10-Core 3.3 GHz (4.1 GHz Turbo) CPU, 128 GB of memory, and NVIDIA RTX2080Ti GPU card.

To evaluate the detection performance of the proposed method, we utilized VGG-net to extract deep-learned features from each face image. Then select a specific layer of the model to describe the features. The first fully connected layer (fc6) has been selected. The criteria



1- Face **2-** Right eye **3-** Left eye **4-** Mouth
5- Correlation between skins

Fig. 9 Illustration of the hand-crafted features

for considering fc6 as a deep-learning feature extractor is selecting the layer that achieves the highest FCM detection accuracy compared to other layers in the VGG model, as shown in Fig. 11. We have used 60% of the dataset as a training set, 20% as a validation set, and 20% as a testing set to evaluate the proposed method's performance. Transfer learning techniques were used to fine-tune the VGG pre-trained network with the Adam optimizer, a learning rate of 0.001, and 50 epochs. Such parameters were selected empirically until optimum performance was achieved.

6.2 Datasets

We conducted experiments in the Celeb-df [2] [26] dataset and further tested the proposed method cross-dataset using different datasets (DFDC[3] [9] and Faceforensics++ [4] [38]). Celeb-df contains 5,639 DeepFake videos generated from 590 real videos on YouTube for

[2] http://www.cs.albany.edu/lsw/celeb-deepfakeforensics.html

[3] https://www.kaggle.com/c/deepfake-detection-challenge

[4] https://github.com/ondyari/FaceForensics

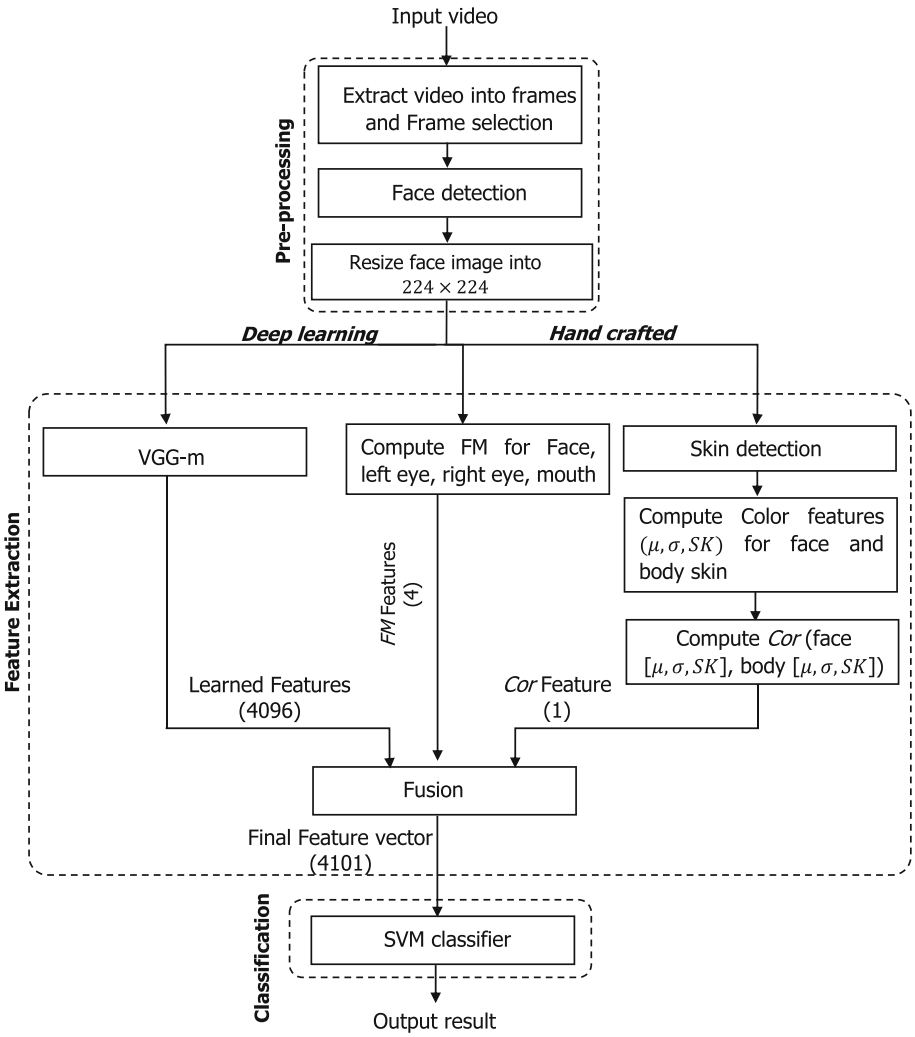


Fig. 10 Workflow digram of the proposed method

59 celebrities of different genders, ages, and ethnic groups. While Faceforensics++ contains 15,000 videos (real and fake) under various compression qualities (i.e., uncompressed data and compressed data with different quantization parameters (0, 23, 40)). Different qualities are used for evaluating the proposed method's robustness at different levels that are shown on several platforms such as Facebook, YouTube, etc. DFDC contains 5,000 videos (real and fake) that Facebook has created. According to the proposed method, all final face images are bilinear-resized to 224×224, which serves VGG baseline resolution. Figure 12 shows examples of the visual results of the proposed method for different datasets. The real faces are marked with green, and the fake faces are marked with red color.

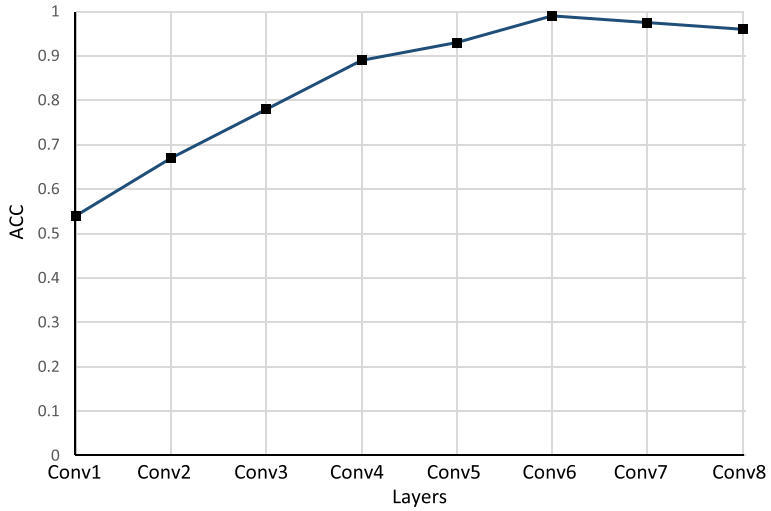


Fig. 11 Classification accuracy based on the selected VGG layers

6.3 Performance metrics

To evaluate the performance of the proposed method, various metrics, i.e., accuracy (Acc), sensitivity (TPR), specificity (TNR), false positive rate (FPR), false negative rate (FNR), predictive values (PPV, NPV), and F-measure, have been used as follows:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

$$TPR = \frac{TP}{TP + FN} \tag{10}$$

$$TNR = \frac{TN}{FP + TN} \tag{11}$$

The false alarm rate, or (FPR) is calculated as $FPR = 1 - TNR$; the miss rate, or (FNR) is calculated as $FNR = 1 - TPR$.

$$PPV = \frac{TP}{FP + TP} \tag{12}$$

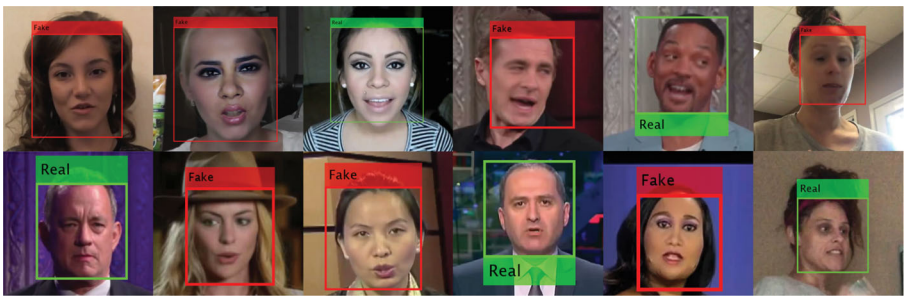


Fig. 12 Some of the visual results of the proposed method

$$NPV = \frac{TN}{FN + TN} \tag{13}$$

$$F - measure = \frac{2PPV \times TPR}{PPV + TPR} \tag{14}$$

If the video is fake and is classified as fake, it is enumerated as a true positive (TP). If the video is real and it is classified as fake, it is enumerated as a false positive (FP). If the video is real and it is classified as real, it is enumerated as a true negative (TN). If the video is fake and it is classified as real, it is enumerated as a false negative (FN).

6.4 Experimental results analysis

Table 1 shows the detection performance of the proposed method on the three different datasets, where the training and testing sets are from the same dataset.

It is noticed that the proposed method achieves high measurements with different datasets based on other manipulation methods. The degree of difficulty in detecting synthesized faces varies with the amount of artifacts appearing and the accuracy of the manipulation method used. Our proposed method achieves the highest performance on the Faceforensics++ dataset because it has a lot of manipulation artifacts that can almost be observed in its videos. The detection task becomes more challenging in the other datasets (Celeb-df and DFDC).

Moreover, the proposed method has been evaluated under different compression quality levels for the videos presented in the Faceforensics++ benchmark. We merged all folders of the same quality level (i.e., C0, C23, C40) and trained the model at each level. The performance of the proposed method has been listed in Table 2.

The proposed method achieves high-performance rates under different compression levels, as shown in the Table 2. However, there are slight effects with C40 (strong compression) due to the manipulation, and artifacts may have relatively vanished. In contrast, the manipulation artifacts tend to be visible under no and light compression.

6.4.1 Robustness test

We present the robustness of the proposed method and the effect of fusing hand-crafted features with deep-learned features. Firstly, we trained the SVM model on the deep-learned features extracted from the VGG network and measured the performance rates on the Celeb-df benchmark. Then, we trained the SVM model on the fused features and remeasured the performance rates. Figure 13 compares the performance rates using only deep-learned

Table 1 Performance of our method on 3 different datasets (Celeb-df, DFDC, and Faceforensics++)

Metrics	Celeb-df	DFDC	Faceforensics++
Acc	0.9954	0.9842	0.9970
TPR	0.9939	0.9902	0.9948
TNR	0.9970	0.9782	1
FPR	0.0030	0.0218	0
FNR	0.0061	0.0098	0.0046
PPV	0.9969	0.9784	1
NPV	0.9939	0.9901	0.9967
F-measure	0.9954	0.9843	0.9970

Table 2 Performance of our method on 3 different compression quality levels in Faceforensics++

Metrics	C0	C23	C40
Acc	0.9993	0.9971	0.9940
TPR	0.9988	0.9949	0.9909
TNR	1	1	1
FPR	0	0	0
FNR	0.0012	0.0054	0.0072
PPV	1	1	1
NPV	0.9989	0.9963	0.9949
F-measure	0.9984	0.9974	0.9953

features and after fusing them with hand-crafted features. It can be seen from the figure that the fusion of hand-crafted features and deep-learned features increases the detection performance of our system.

6.5 Comparison with the state-of-art methods

We compare our method with the hand-crafted-based methods in [28] (where the texture features were extracted from video frames) and [43] (where temporal motion was used). Moreover, the proposed method is compared with FCM detection methods based on deep learning such as [1, 15, 20, 23, 32, 39, 44, 47]. To make a fair comparison, we fine-tuned the networks used in the previous methods on the three datasets (Celeb-df, DFDC, and Faceforensics++), and the performance rates have been evaluated. Table 3 lists the detection accuracy of the proposed method against the state-of-the-art methods. Our method improved detection accuracy by about 1.89% on Celeb-df, 2.55% on DFDC, and 0.6% on the Faceforensics++ benchmark.

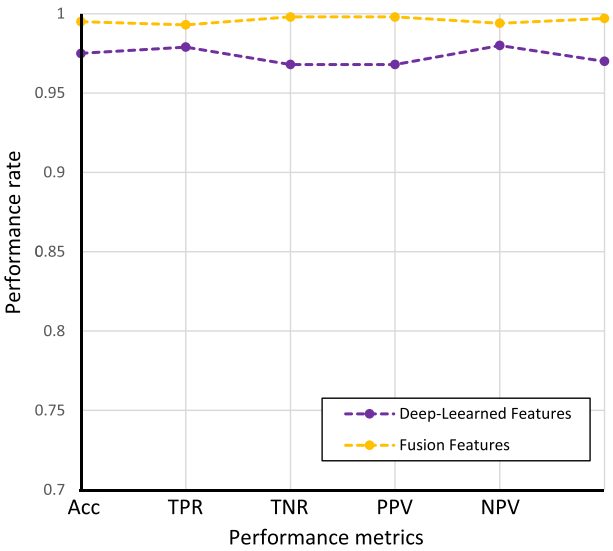


Fig. 13 The performance comparison between using only deep-learned features and using fusion features

Table 3 Comparison of the proposed method and the state-of-art methods

Methods	Celeb-df	DFDC	Faceforensics++
Matern [28]	0.8342	0.7956	0.8604
Wang [43]	0.9769	0.9597	0.9910
Afchar [1]	0.9322	0.9168	0.9526
Kumar [23]	0.9543	0.9223	0.9739
Sabir [39]	0.9501	0.9312	0.9672
Wu [44]	0.9400	0.9047	0.9533
khalid [20]	0.8796	0.8623	0.9030
Zhou [47]	0.9456	0.9278	0.9636
Megahed [32]	0.9302	0.9203	0.9788
Fung [15]	0.9000	0.8902	0.9300
Ours	0.9954	0.9842	0.9970

7 Discussion

According to the results yielded in Table 3, it is evident that the proposed method has a distinct detection performance among the state-of-the-art methods. Our method combines the advantages of hand-crafted and deep-learning feature extractors, where the proposed hand-crafted features focus on exposing specific deepfake artifacts (i.e., obliteration, blurriness, color intensity, and illumination). In contrast, the pre-trained VGG model automatically extracts deep features from the face image. The fusion of hand-crafted and deep-learning features can improve classification accuracy, as shown in Fig. 13. Furthermore, Table 1 shows that the proposed method performed well with different datasets regardless of the manipulation method used.

The main advantages of the proposed method are summarized as follows:

1. The proposed method improved the detection performance using a fusion of hand-crafted and deep-learned features.
2. Using FM greatly assists in extracting the amount of blurriness from face images, and the correlation between face skin and body skin helps to measure the similarity, which leads to an increase in the efficiency of the proposed method.
3. Implementing the proposed method is accessible and achieves superior results compared to state-of-art methods.
4. The proposed method achieves high performance across different datasets.

According to the previous advantages of the proposed method, it can be deployed in real-world applications as a very competitive method. However, with the stunning and continuous development of deepfake generation methods, the manipulation artifacts may shrink, affecting detection accuracy.

8 Conclusion

DeepFake tools and software recently invaded social media and smartphones, leading to distrust of visual content. Creating fake news to distort politicians, celebrities, or anyone else became effortless. This leads to destabilization, making an urgent need to investigate visual content. In this paper, we utilized a fusion technique to fuse learned features with

hand-crafted features to increase detection performance. The VGG network is used to extract the deep-learned features from the convolutional base of the model. Then fuse these features with other hand-crafted features (amount of blurriness and similarity between face and body skin). The experimental results on various datasets demonstrated that the proposed fusion model performs better than the other state-of-the-art methods. In the future, we intend to make our system verify audio content as well, which could provide a significant improvement.

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Data Availability The datasets used and/or analyzed during the current study are available in the Celeb-DF, DFDC, and FaceForensics++ repositories respectively,
http://www.cs.albany.edu/lsw/celeb-deepfakeforensics.html
https://www.kaggle.com/c/deepfake-detection-challenge
https://github.com/ondyari/FaceForensics

Declarations

Conflicts of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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